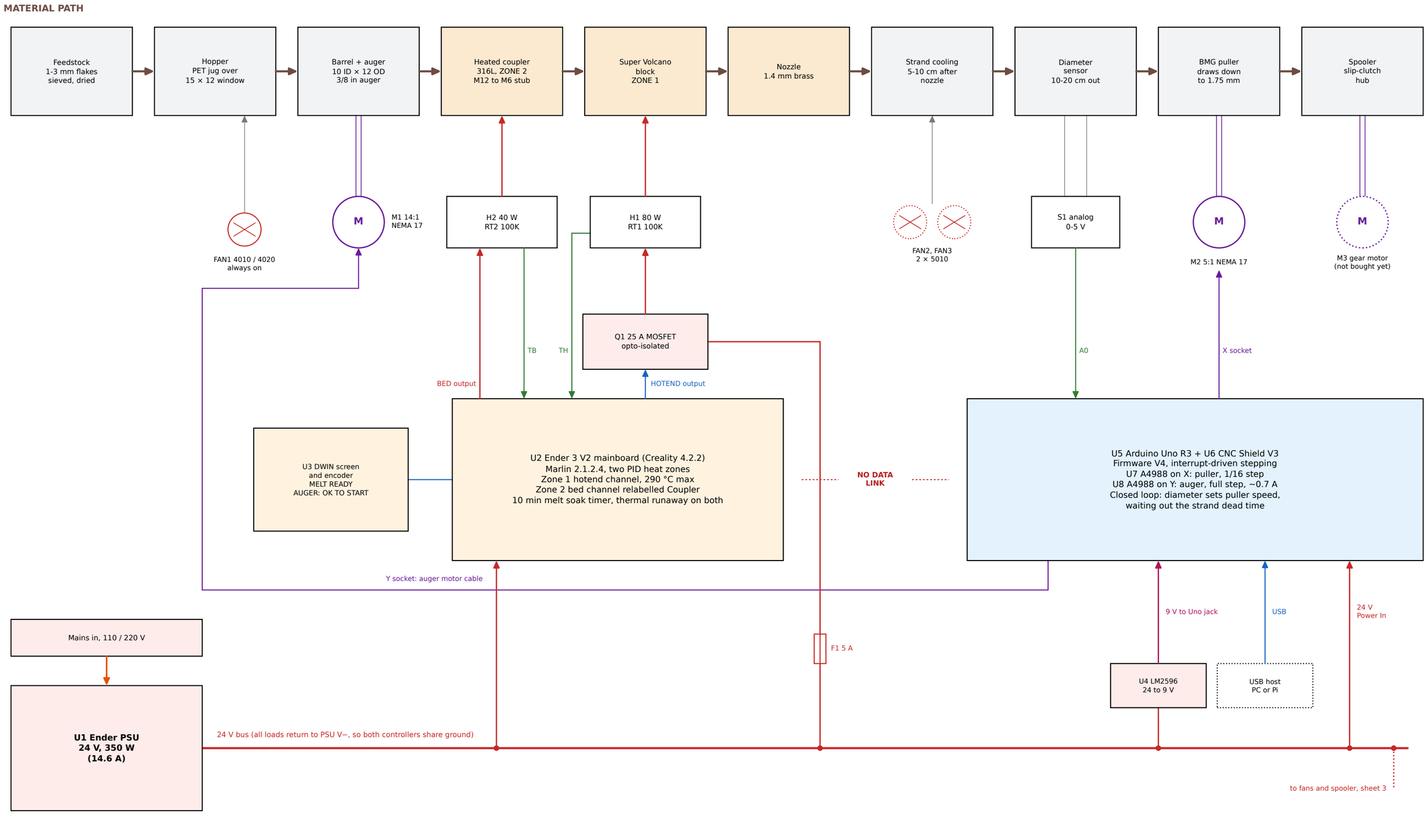
System overview

Plastic runs left to right along the top. Two independent controllers run the line: the Ender mainboard owns heat, the Arduino owns motion.



Heater and motor power

Stepper drive

Control signal

Material flow

Temperature / diameter sense

Not specified in the docs

The Arduino cannot see temperature. Starting the auger only after AUGER: OK TO START is an operator step, not an electrical interlock.

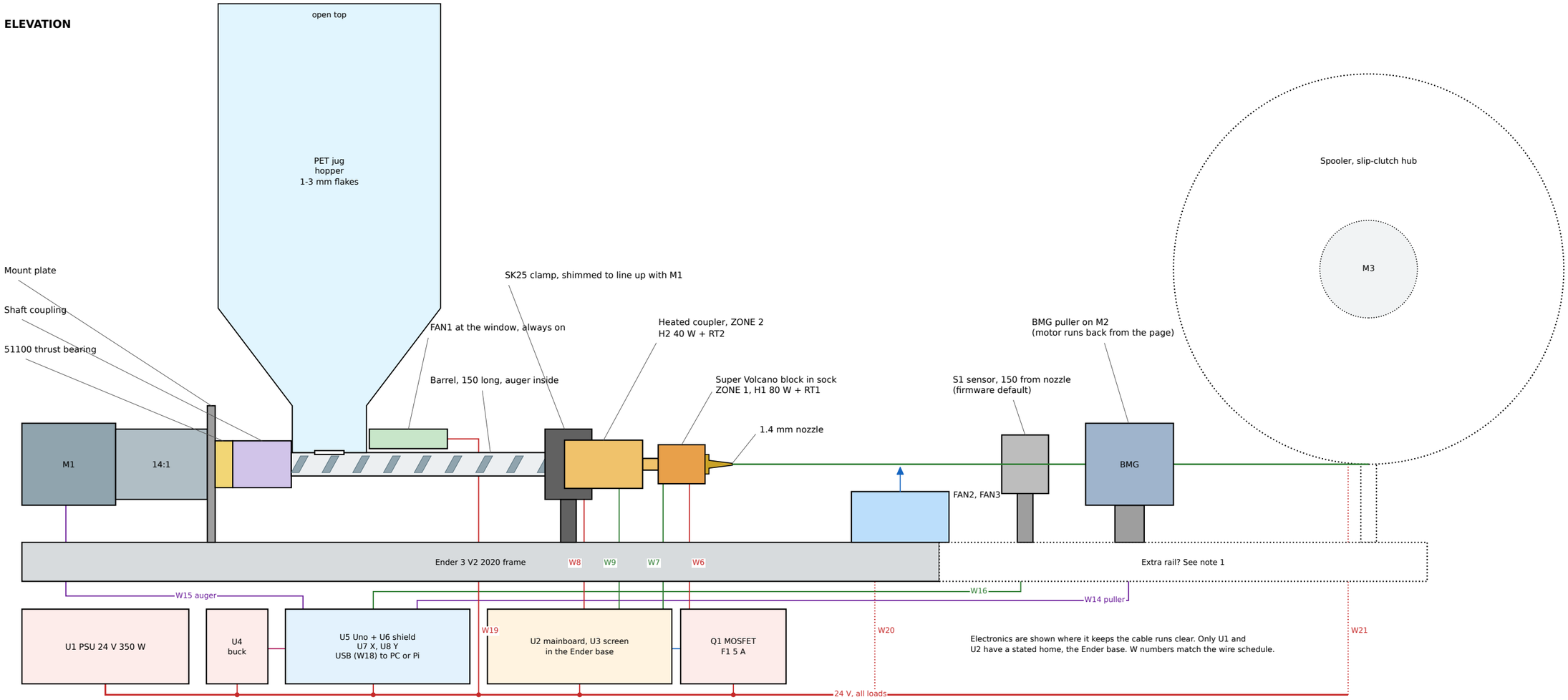
Details: sheet 2 assembled device, sheet 3 power and heaters, sheet 4 controller, sheet 5 mechanical, sheet 6 wire schedule.

3DTruder auger-fed filament extrusion line		
System overview		
Drawn from the build docs and firmware in this repo	Rev B 2026-11	**Sheet 1 of 6**

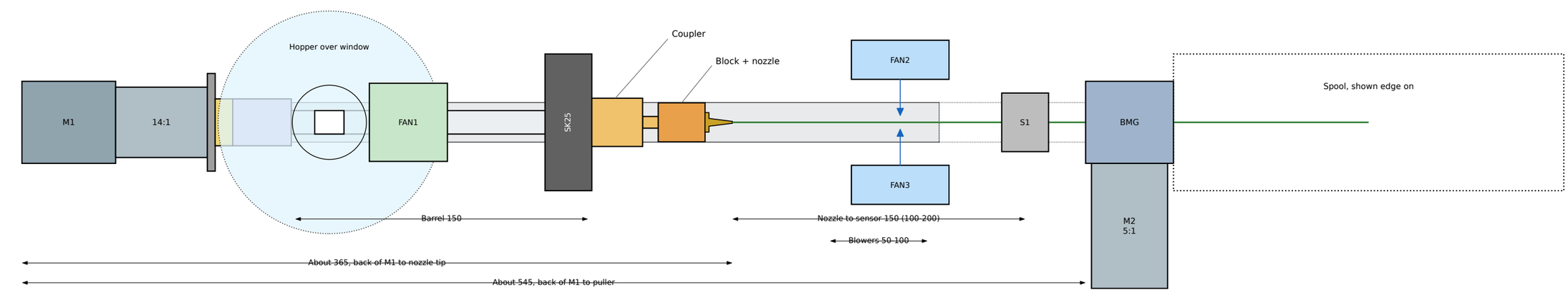
Fully assembled device

Every part in its built position at one common scale (about 1:2.25 at A3), with the electronics and cable runs. Heights and lengths of bought parts are typical values.

ELEVATION



PLAN

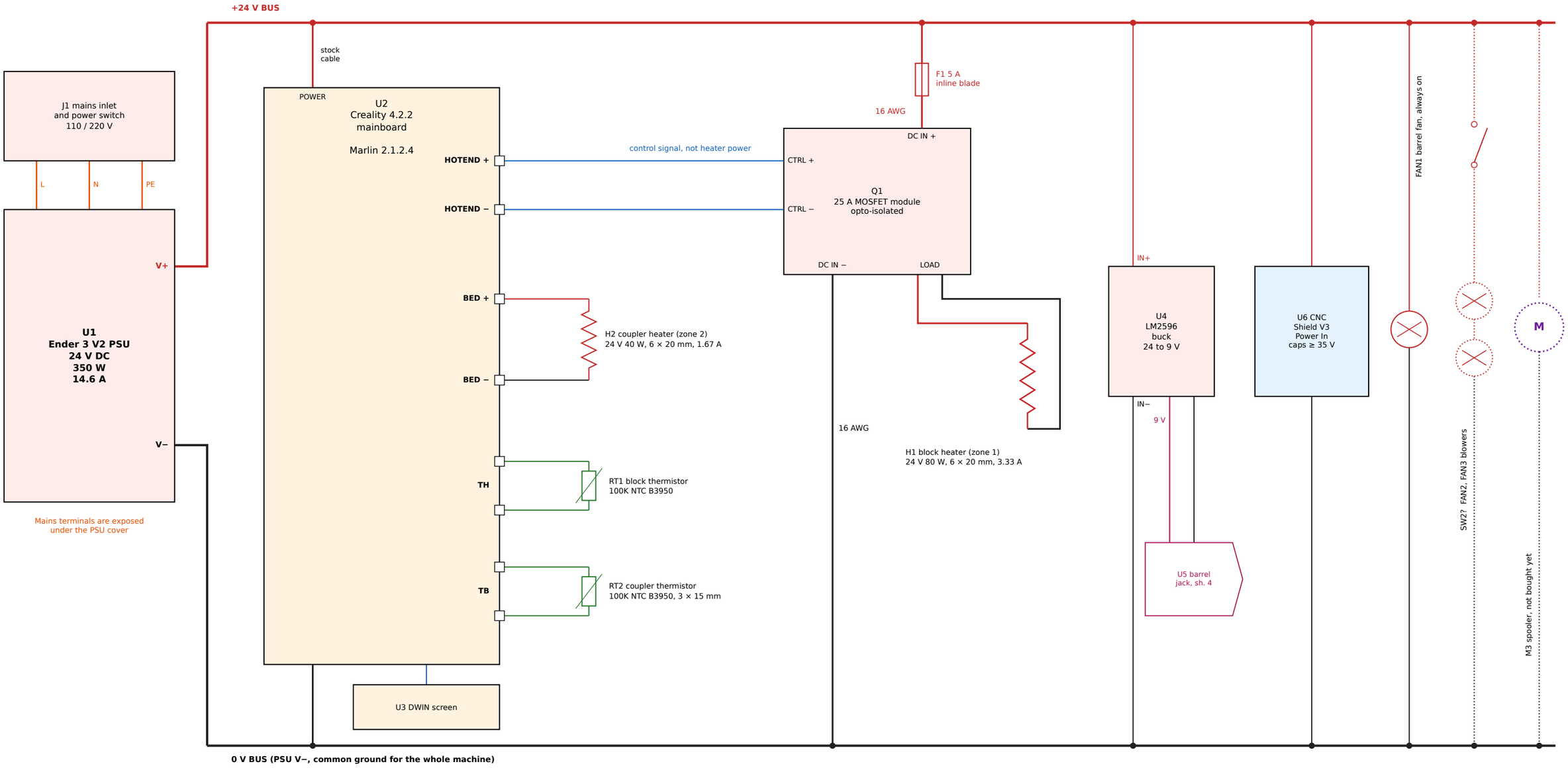


1. The docs say the line bolts to the 2020 frame but not to which rail. Back of M1 to puller is about 545 mm and the Ender's footprint is roughly 470 mm, so check the fit. The puller and spooler may need an extra rail or their own stand. Dotted frame is that open question.
2. M1, M2, FAN, BMG, SK25 and spool sizes are typical, not from the repo. Coupler, barrel, thrust bearing and spacings are from the docs. Ender gantry, motors and belts are not drawn.

3DTruder auger-fed filament extrusion line		
Fully assembled device		
Drawn from the build docs and firmware in this repo	Rev B 2026-11	**Sheet 2 of 6**

Power distribution and heaters

Ladder layout: the 24 V bus runs along the top, the 0 V bus along the bottom, and every load is a rung between them.



Notes

1. Unplug from the wall before touching PSU or mainboard wiring.
2. H1 must go through Q1. Its 3.33 A is more than the mainboard MOSFET should carry continuously. H2's 1.67 A is within the bed MOSFET's rating.
3. Connect both thermistors before heating. An unplugged one halts that zone with MINTEMP (5 °C). Limits: nozzle MAXTEMP 305 °C (290 °C settable), coupler 290 °C (280 °C settable).

Open wiring

1. Dotted rungs: the docs do not say how FAN2, FAN3 or M3 are powered. The bus feed and SW2 are the obvious way to do it, not a documented design. A 12 V spooler needs its own buck.

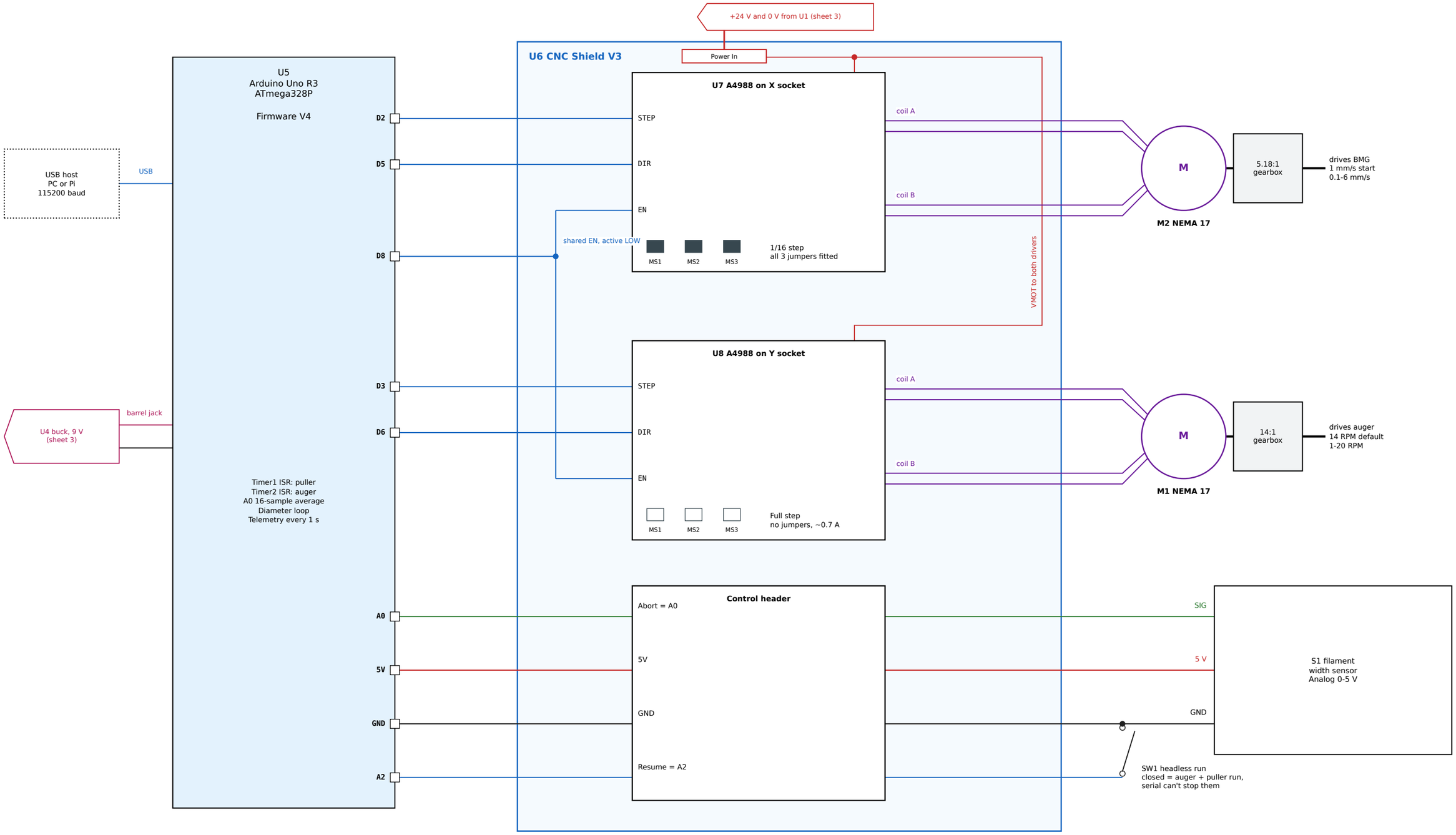
Mains · +24 V · 0 V · Control signal

Thermistor · 9 V · Not specified in the docs

3DTruder auger-fed filament extrusion line		
Power distribution and heaters		
Drawn from the build docs and firmware in this repo	Rev B 2026-11	**Sheet 3 of 6**

Motion controller: Arduino Uno and CNC Shield V3

U6 plugs straight onto U5, so the links between them are header pins, not loose wires. Only the connections at the far left and right are wired by hand.



1. Check the shield's capacitors before applying 24 V: 35 V or 50 V only. Budget shields often ship with 16 V or 25 V parts. Otherwise replace them or run the shield from 12 V.
2. Set U4's output to 9 V with a multimeter before connecting it to the Uno. These modules are adjustable and can arrive set well above 9 V.
3. Auger current is the torque limit: a jam must make M1 skip, not build pressure against the thin barrel thread. Don't raise it to push through a jam.

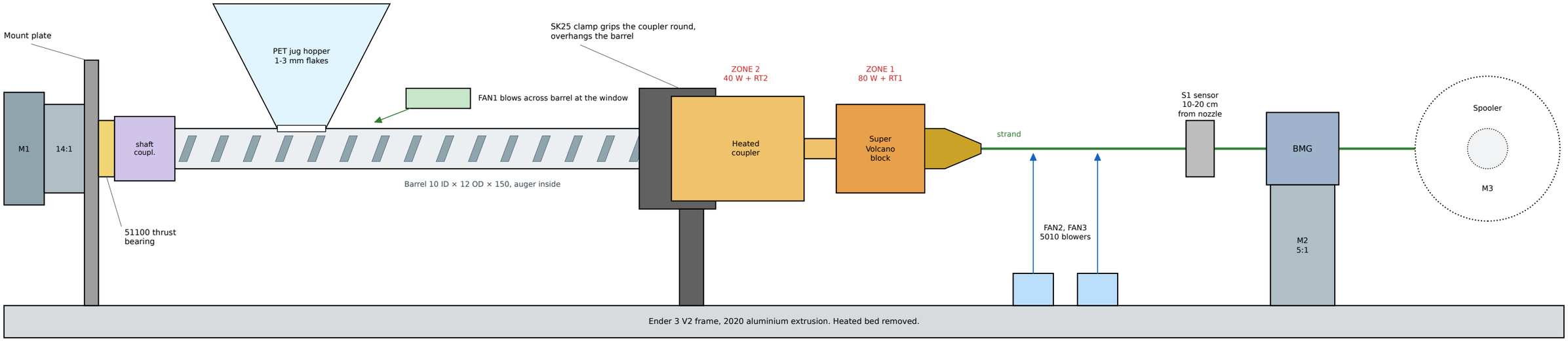
Step / dir / enable — Stepper coils — Sensor signal — Supply

3DTruder auger-fed filament extrusion line		
Motion controller: Arduino Uno and CNC Shield V3		
Drawn from the build docs and firmware in this repo	Rev B 2026-11	**Sheet 4 of 6**

Mechanical layout and melt path

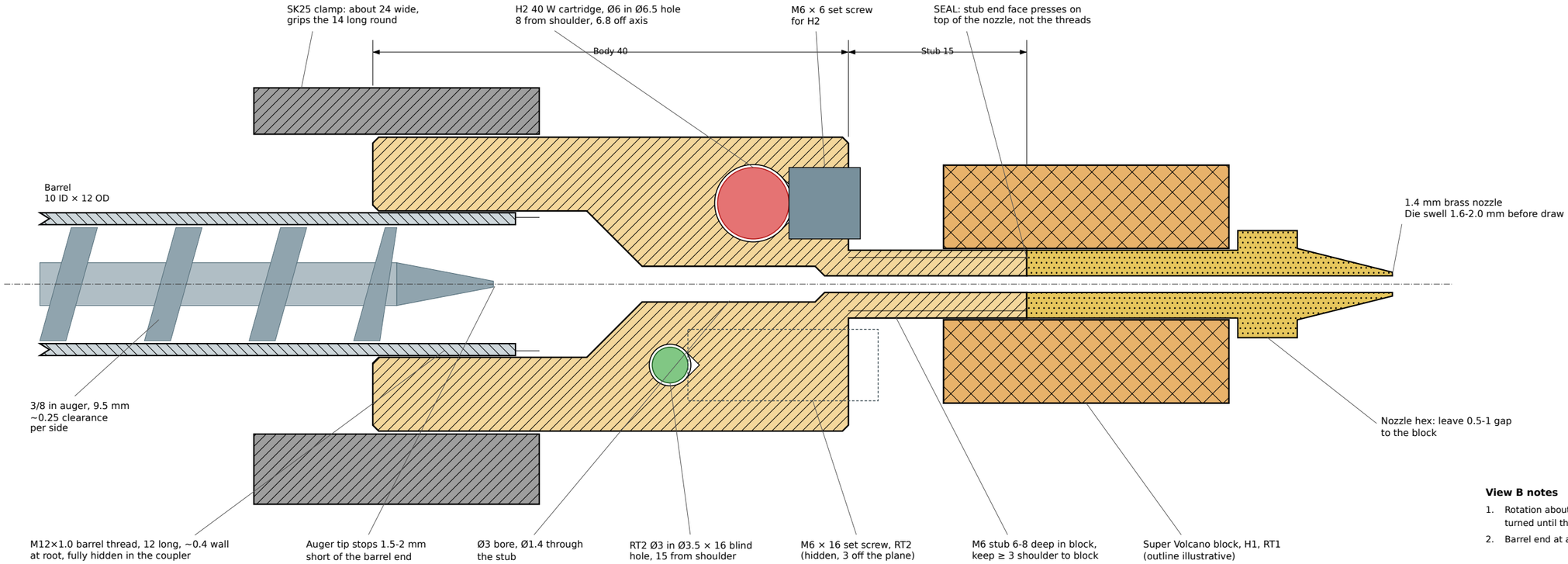
View A is a side view of the whole line, not to scale. View B is a section through the melt path at 2.5:1, with coupler and barrel to scale.

VIEW A Full line, side view (not to scale)



Load paths: melt pressure on the auger goes coupling → 51100 → plate → frame, not through the gearbox. Pressure on the coupler goes SK25 → frame, not through the thin barrel thread.

VIEW B Melt path section, 2.5:1



View B notes

1. Rotation about the axis is set by the barrel thread, then the whole assembly is turned until the hopper window faces up. Drawn with H2 on top for clarity.
2. Barrel end at about z 28, assuming the 12 mm thread is fully run in.

3DTruder auger-fed filament extrusion line		
Mechanical layout and melt path		
Drawn from the build docs and firmware in this repo	Rev B 2026-11	**Sheet 5 of 6**

Wire schedule and settings

Every connection and every value you need at the bench. Source for each is the manual, the sketch or Configuration.h.

Wire schedule

Wire	From	To	Carries	Cable / note
W1	J1 mains inlet	U1 PSU L, N, PE	110 / 220 V AC	Stock. Unplug before touching
W2	U1 V+, V−	U2 POWER	24 V	Stock cable
W3	U1 V+	F1, then Q1 DC IN +	24 V, 3.33 A	16 AWG red, 5 A fuse on +
W4	U1 V−	Q1 DC IN −	0 V	16 AWG black
W5	U2 HOTEND +, −	Q1 CONTROL +, −	PWM control	Signal cable, mind polarity
W6	Q1 LOAD	H1 80 W block heater	24 V, 3.33 A	Heater leads
W7	RT1 block thermistor	U2 TH	Temp sense	Marlin sensor type 1
W8	H2 40 W coupler heater	U2 BED screw terminals	24 V, 1.67 A	Route away from nozzle
W9	RT2 coupler thermistor	U2 TB	Temp sense	Marlin sensor type 1
W10	U2	U3 DWIN screen	Serial UI	Stock cable
W11	U1 V+, V−	U6 Power In	24 V motor power	Check caps first
W12	U1 V+, V−	U4 IN	24 V	
W13	U4 OUT	U5 barrel jack	9 V	Set 9 V before connecting
W14	U6 X motor header	M2 puller, 5:1	Stepper coils	Swap a coil pair to reverse
W15	U6 Y motor header	M1 auger, 14:1	Stepper coils	Swap a coil pair to reverse
W16	U6 Abort (A0), 5V, GND	S1 SIG, VCC, GND	0-5 V analog	Not a digital runout switch
W17	U6 Resume (A2), GND	SW1 run switch	Closed = run	Optional, headless only
W18	U5 USB	PC or Pi	115200 baud	Commands and telemetry
W19	U1 V+, V−	FAN1 4010 / 4020	24 V	Always on, no switch
W20	Not specified	FAN2, FAN3 5010	24 V assumed	OPEN: docs give no supply or switch
W21	Not specified	M3 spooler	24 V or 12 V	OPEN: not bought. 12 V needs a buck

Drivers

Stepper driver	Socket / jumpers	Step mode	Current
U7 puller (M2)	X, 3 jumpers fitted	1/16	Motor's comfortable running current
U8 auger (M1)	Y, no jumpers	Full, 2800 steps/rev	~0.7 A, torque limit

Vref = 8 × Rs × I. At 0.7 A: R100 sense resistor 0.56 V, R050 sense resistor 0.28 V.

Presets

Preset	Nozzle	Coupler
PLA	215 °C	205 °C
PETG	240 °C	230 °C
PET #1	270 °C	260 °C

Arduino pin map

Arduino pin	Shield label	Function
D2 (PD2) / D5	X STEP / DIR	Puller, Timer1 ISR
D3 (PD3) / D6	Y STEP / DIR	Auger, Timer2 ISR
D8	EN	Active LOW, both drivers
A0	Abort	Diameter sensor
A2	Resume	Run switch to GND, pull-up

Thermal protection

Marlin setting	Value
Soak timer	10 min, within 3 °C, resets past 5 °C
MINTEMP both zones	5 °C
MAXTEMP nozzle / coupler	305 / 290 °C
Settable max	290 / 280 °C
Thermal runaway	60 s, 4 °C, both zones

Calibration

Controller value	Default	Set at commissioning
PULL_STEPS_PER_MM	2150	2150 × 60 / L
SENSOR_DISTANCE_MM	150	Measure nozzle to sensor
CAL_ADC_A at 1.50 mm	300	Read RAW with gauge pin
CAL_ADC_B at 2.00 mm	700	Read RAW with gauge pin
TARGET_DIA / dead band	1.75 / ±0.02	Fixed
Correction	Gain 0.5	Capped ±10 %, min 2 s apart

Serial, 115200 baud, Newline

Serial command	Action
START / STOP, OFF	Auger + puller on / off
PULL ON / OFF	Puller only, for threading
AUG ON / OFF	Auger only, 1.5 s soft start
SPEED <x>	Puller, 0.1-6 mm/s
RPM <x>	Auger, 1-20 RPM
AUTO ON / OFF	Diameter regulation on / frozen
HELP	List commands

Open items at Rev B

1. FAN2, FAN3 supply and switching (W20). The procedure says "Blowers on" but gives no wiring, and the screen menu has no fan item.
2. M3 spooler (W21) is still to buy, so its voltage is not settled.
3. F1 protects only the heater branch. U4, U6, FAN1 and the blower and spooler rungs have no fuse of their own in the docs.
4. The barrel's M12×1.0 thread is listed as still to do in the parts list.

Reference designators

1. U1 PSU, U2 mainboard, U3 screen, U4 buck, U5 Uno, U6 shield, U7 U8 A4988, Q1 MOSFET, F1 fuse, H1 H2 heaters, RT1 RT2 thermistors, M1 auger, M2 puller, M3 spooler, FAN1 barrel fan, FAN2 FAN3 blowers, S1 diameter sensor, SW1 run switch, SW2 blower switch (assumed), J1 mains inlet.

Power and unused hardware

1. Known heater load is 120 W, or 5.0 A of the PSU's 14.6 A. Motors and fans aren't rated in the docs, but that leaves plenty of margin.
2. Ender X, Y, Z and E motor outputs and endstops are not used by this build.

Bring-up order

1. Before any power: check the shield's capacitors are 35 V or 50 V, and set U4 to 9 V on a multimeter with the Uno disconnected.
2. Flash Marlin to U2 and the sketch to U5. Set Vref on U7 and U8 and fit the jumpers.
3. First boot, cold, both thermistors plugged in: the nozzle and COUPLER readouts should both show 20-25 °C.
4. Heat to 250 °C, hold the coupler flats with a 22 mm wrench and tighten the nozzle a further 1/8 turn against the stub.
5. PID tune both zones and save: M303 E0 S220 C8 U1, then M303 E-1 S205 C8 U1, then M500.
6. Heaters off, barrel empty: check both motor directions, calibrate S1 with gauge pins, then puller steps/mm and sensor distance. Re-upload the sketch.
7. Watch one full soak on a preset, through SOAKING: 10:00 to AUGER: OK TO START, before the first run.